COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S3
DATE PERFORMED:	7/9/26	**DATE SUBMITTED:**	7/11/26
NAME:	SAMUEL D. RAMOS	**INSTRUCTOR:**	ENGR. JIMLORD M. QUEJADO

ACTIVITY NO. __1_ [Hands-on Activity 1.1 Basic C++ Programming]

1. PROCEDURE OUTPUT

ILO A

```cpp
// ILO A

#include <iostream>

int sumOfTwoNum(int fNum, int sNum);
int isGreater(int fNum, int sNum);
bool logicalOperators(bool a, bool b);

int main(){


    std::cout << "The sum of two numbers is: " << sumOfTwoNum(6,5) << std::endl;
    isGreater(6,5);

    if (logicalOperators(true, false)){
        std::cout << "Success!" << std::endl;
    }


return 0;

}

int sumOfTwoNum(int fNum, int sNum){

    int sum = fNum + sNum;

```

```
28      return sum;
29  }
30
31  int isGreater(int fNum, int sNum){
32
33      if (fNum > sNum){
34          std::cout << "The first number is greater than the second number." << std::endl;
35      }
36      else{
37          std::cout << "The first number is less than the second number." << std::endl;
38      }
39  }
40
41  bool logicalOperators(bool a, bool b){
42
43      std::cout << "A&&B = " << (a&&b) << std::endl;
44      std::cout << "A||B = " << (a||b) << std::endl;
45      std::cout << "!A = " << (!a) << std::endl;
46      std::cout << "!B = " << (!b) << std::endl;
47
48      return true;
49  }
50
```

OUTPUT

```
The sum of two numbers is: 11
The first number is greater than the second number.
A&&B = 0
A||B = 1
!A = 0
!B = 1
Success!
```

ANALYSIS

- For the first function, it simply gets two integers, adds it, then prints out the sum. For the second function, it checks whether the first input is greater than the 2nd input, it will print out what it discovers, if the first is greater or not. For the third function, it takes two boolean values (true and false) then uses all the logical operators then return true. This will then display a confirmation message.

ILO B

```cpp
// ILO B

#include <iostream>

// naming convention for class name: PascalCase (first letter is always capital)
class Triangle{
    private:
        double totalAngle,angleA,angleB,angleC;

    public:
        // constructor
        Triangle(double A, double B, double C);
        void setAngels(double A, double B, double C);
        const bool validateTriangle();

};



int main(){

    Triangle set1(40,30,110);

    if (set1.validateTriangle()){
        std::cout << "The shape is a valid triangle \n";
    }

```

```
28      else{
29          std::cout << "The shape is NOT a valid triangle \m";
30      }
31
32      return 0;
33
34  }
35
36
37
38  Triangle::Triangle(double A, double B, double C){
39      angleA = A;
40      angleB = B;
41      angleC = C;
42      totalAngle = A+B+C;
43  }
44
45  void Triangle::setAngels(double A, double B, double C){
46      angleA = A;
47      angleB = B;
48      angleC = C;
49      totalAngle = A+B+C;
50  }
51
52  const bool Triangle::validateTriangle(){
53      return(totalAngle<=180);
54  }
55
```

OUTPUT

```
The shape is a valid triangle

--------------------------------
Process exited after 0.3248 seconds with return value 0
Press any key to continue . . .
```

ANALYSIS

- This code uses a class which has a public and a private. The private is manipulated through the use of the public, by setting values to the public variables it sets angles, and then it checks whether the angle is valid for a triangle or not.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C #1

```cpp
// ILO C1

#include <iostream>


int main(){

    int num1 = 5;
    int num2 = 6;
    int temporaryNum;

    std::cout << "First number: " << num1 <<  std::endl;
    std::cout << "Second number: " << num2 << std::endl;

    temporaryNum = num2;
    num2 = num1;
    num1 = temporaryNum;

    std::cout << "First number: " << num1 <<  std::endl;
    std::cout << "Second number: " << num2 << std::endl;

    return 0;
}
```

OUTPUT

```
First number: 5
Second number: 6
First number: 6
Second number: 5

--------------------------------
Process exited after 0.3125 seconds with return value 0
Press any key to continue . . .
```

ANALYSIS

- This code swaps two numbers in different variables by keeping the value of the 2nd number in a temporary variable, which then the 2nd takes the value of the 1st number and then the 1st number takes the value of the 2nd number stored in the temporary variable.

ILO C#2

```cpp
// ILO C2

#include <iostream>

float kelvinToFahrenheit(float kelvin);

int main(){

    float kelvin = 0;
    std::cout << "Enter the temperature in Kelvin: " << kelvin << std::endl;


    std::cout << "Kelvin converted to Fahrenheit is: " << kelvinToFahrenheit(kelvin);

    return 0;
}

float kelvinToFahrenheit(float kelvin){
    float fahrenheit = (kelvin - 273.15)*(9.0/5.0) + 32;

    return fahrenheit;
}

```

OUTPUT

```
Enter the temperature in Kelvin: 0
Kelvin converted to Fahrenheit is: -459.67
--------------------------------
Process exited after 0.3296 seconds with return value 0
Press any key to continue . . .
```

ANALYSIS

- The function receives the value in kelvin, using the formula for transitioning kelvin to fahrenheit, the value of kelvins is then turned to fahrenheit then displayed.

ILO C#3

```cpp
// ILO C #3
#include <iostream>


int distance(int x, int y);

int main(){

    std::cout << "The distance between the points 135 and 200 is: " << distance(135,200) << std::endl;

    return 0;
}

int distance(int x, int y){
    int d;
    d = y - x;

    return d;
}
```

OUTPUT

```
The distance between the points 135 and 200 is: 65


--------------------------------
Process exited after 0.3164 seconds with return value 0
Press any key to continue . . .
```

ANALYSIS

- The function checks the distance between point 1 and point 2 by subtracting point 2 by point 1 then returning its value.

ILO C#4

```cpp
// ILO C#4
// modified ilo B
#include <iostream>
#include <cmath>


class Triangle{
    private:
        double totalAngle,angleA,angleB,angleC;

    public:
        // constructor
        Triangle(double A, double B, double C);
        void setAngels(double A, double B, double C);
        const bool validateTriangle();

};

float areaOfTriangle(float base, float height);
int perimeterOfTriangle(int a, int b, int c);
void angleOfTriangle(int a, int b, int c);


int main(){

    Triangle set1(40,30,110);

    if (set1.validateTriangle()){
        std::cout << "The shape is a valid triangle \n";
```

```
    }

    else{
        std::cout << "The shape is NOT a valid triangle \n";
    }

    std::cout << "The area of a triangle with base of 11 and height of 20 is: " << areaOfTriangle(11,20) << std::endl;

    std::cout << "The perimeter of a triangle with the sides 5, 8 ,7 is: " << perimeterOfTriangle(5,8,7) << std::endl;

    std::cout << "The angle of the triangle with the sides 5,8,7 is: ";
    angleOfTriangle(5,8,7);

    return 0;

}



Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
void Triangle::setAngels(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

const bool Triangle::validateTriangle(){
    return(totalAngle<=180);
}

float areaOfTriangle(float base, float height){
    return 1.0/2.0*(base*height);
}

int perimeterOfTriangle(int a, int b, int c){
    return a+b+c;
}

void angleOfTriangle(int a, int b, int c){
    int temporary;
    int e = 2;

    if (a>b){
        temporary = a;
        a = b ;
        b = temporary;
    }
```

]

```
84
85      if (b>c){
86          temporary = b;
87          b = c;
88          c = temporary;
89      }
90
91      if (pow(a,e)+pow(b,e)==pow(c,e)){
92          std::cout << "Right triangle" << std::endl;
93      }
94
95      else if (pow(a,e)+pow(b,e)>pow(c,e)){
96          std::cout << "Acute triangle" << std::endl;
97      }
98
99      else{
100         std::cout << "Obtuse triangle" << std::endl;
101     }
102 }
103
```

OUTPUT:

```
The shape is a valid triangle
The area of a triangle with base of 11 and height of 20 is: 110
The perimeter of a triangle with the sides 5, 8 ,7 is: 20
The angle of the triangle with the sides 5,8,7 is: Acute triangle

--------------------------------
Process exited after 0.3327 seconds with return value 0
Press any key to continue . . .
```

ANALYSIS

- The function to get the area of the triangle stores the value of a triangle's base and height then returns the computed area. The function to get the perimeter of a triangle stores a triangle's three sides, and then computes its perimeter by returning the sum of all three sides of the triangle. The function which determines what angle the triangle is works by using the pythagorean theorem. The important step is to find the largest number and set it to the variable c, which is found by using if statements to check for the largest value and swap it with c.

CONCLUSION

- With this activity, I have recalled how to code using the language c++. Although I haven't recalled all the techniques I used in the past, I can still confidently learn CPE010 (Data Structures and Algorithm). There are still more I have to recall, and I'm sure it will be of great help in taking this course.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Formulas, and formatting**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Formula of area, kelvin to fahrenheit, perimeter, how to check the angle of a triangle, and how to set up boolean logical operators.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.